

Ritika Kalia

Computer Science & Engineering

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EDUCATION

Bachelor of Engineering in Computer Science & Engineering	2024 – 2027
Chandigarh College of Engineering & Technology	CGPA: 9.45/10 (till 6th sem)
Diploma in Computer Science & Engineering	2022 – 2024
Chandigarh College of Engineering & Technology	86%
Intermediate / Class XII	2020 – 2021
Shishu Niketan Model Sr. Sec. School	91.8%
Matriculation / Class X	2018 – 2019
Shishu Niketan Model Sr. Sec. School	95%

TECHNICAL SKILLS

Programming Languages: Python, C, SQL, JavaScript

Web Development: HTML, CSS, Django, Flask, Streamlit

Deep Learning: PyTorch, Hugging Face Transformers

Concepts: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems

PROJECTS

Graph Algorithm Visualizer Demo | GitHub

JavaScript, SVG, HTML/CSS, Django

- Built an interactive BFS/DFS visualizer with play/pause, step-forward and step-back playback, served via Django.
- Organized the frontend into core (graph data), algorithm (step-generators), and UI (rendering, playback) layers using modular JavaScript.
- Synchronized live **Queue** (BFS) and **Stack** (DFS) panels with a step-by-step explanation log of algorithm actions.
- Supported directed, undirected and weighted edges, drag-and-drop nodes, preset graphs, zoom/pan, touch and keyboard controls, and a resizable canvas up to **4000×3000**.

CPU Scheduling Simulator – Visualizing OS Scheduling Algorithms Demo | GitHub

Python, Streamlit, Matplotlib, Pandas

- Implemented **8 CPU scheduling algorithms** including FCFS, SJF, SRTN, Round Robin, LJF, LRTN, and Static/Dynamic Priority with CPU + I/O mixed workloads, preemption, and aging-based starvation prevention.
- Added **4 workload sources**: preset CPU-only, mixed CPU+I/O, random generator, and manual input, with a Compare-All mode for evaluating all algorithms on the same workload.
- Visualized scheduling results using dual Matplotlib Gantt charts for CPU and I/O along with **6 metrics**: completion, waiting, turnaround, response, CPU utilization, and throughput.

Full-Stack E-Commerce Platform – StellerCart Demo | GitHub

Python, Django, Bootstrap, HTML, CSS, JavaScript

- Built a Django e-commerce application with email/password and Google OAuth login, product search, reviews, cart, checkout, and order history using **8 models, 19 routes, and 10 templates**.
- Integrated **Razorpay** in test mode with server-side amount calculation and HMAC signature verification, alongside Cash on Delivery.
- Implemented server-side calculation of subtotal, **10% tax**, and shipping with decimal rounding; order history stores a snapshot of the shipping address at order time.
- Validated Indian shipping details including mobile number, PIN code, city, and state using a local **India Post PIN directory** with API fallback; added AJAX delivery updates with CSRF protection.

EXPERIENCE

CS Soft Solutions (I) Pvt. Ltd. Jul 2023 – Aug 2023

Python Programming – Industrial Training

- Completed industrial training in Python programming fundamentals and applied scripting.

RESEARCH AND PUBLICATIONS

IEEE ICIR 2026 – Research Publication & e-Poster Presentation

Real-Time Privacy Guardianship in XR Metaverses via AI-Driven Behavioral Analytics

- Developed a behavioral analytics pipeline for detecting VR privacy attacks, achieving **92% accuracy** and **0.976 ROC-AUC**.
- Presented the research as an e-poster at the University of Pisa, Italy.
- IEEE Xplore | GitHub

LEADERSHIP AND ACHIEVEMENTS

- Chairperson**, ACM-W CCET Student Chapter, 2025.
- Technical Team Member**, CCET APRATIM 2025 College Tech Fest.
- Design Contributor**, CCET College Website Team; designed official departmental web pages.
- 3rd Position**, Business Blueprint 2025, IIC CCET; pitched a multimodal AI/ML Women Safety Analytics System.